

11. Rating Political Candidates

A political scientist is studying the relationship between the voter image of a conservative political candidate and the distance (in kilometers) between the residences of the voter and the candidate. Each of 12 voters rated the candidate on a scale of 1–20.

Voter	Rating	Distance	d_i	d_i^2
1	12	75	3	9
2	7	165	1	1
3	5	300	12	144
4	19	15	1	1
5	17	180	8	64
6	12	240	11	121
7	9	120	4	16
8	18	60	2	4
9	3	230	10	100
10	8	200	9	81
11	15	130	5.5	30.25
12	4	130	5.5	30.25

- Calculate Spearman's rank correlation coefficient r_s .
- Do these data provide sufficient evidence to indicate a negative rank correlation between rating and distance?

$$n = 12, \sum d^2 = 454$$

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} = 1 - \frac{6(454)}{12(12^2 - 1)} = -0.5874$$

$\therefore$ Rating, Distance 負相關
遠 $\rightarrow$ 評分 $\downarrow$

$$b, \text{ 設 } \alpha = 0.05$$

$$\begin{aligned} H_0: \rho_s &= 0 \\ H_a: \rho_s &< 0 \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{左單尾}$$

$$n = 12, \alpha = 0.05$$

$$r_{s, \text{critical}} \approx -0.503$$

$$\text{實際計算結果 } r_s = -0.5874 < -0.503$$

$\therefore$ reject H_0 , 有 sufficient evidence to show Rating, Distance 之間的確是負相關

12. Competitive Running Is the number of years of competitive running experience related to a runner's distance running performance? The data on nine runners, obtained from a study by Scott Powers and colleagues, are shown in the table:⁸

Runner	Years of Competitive Running	10-Kilometer Finish Time (minutes)	d_i	d_i^2
1	9	33.15	1	1
2	13	33.33	2	4
3	5	33.50	3	9
4	7	33.55	4	16
5	12	33.73	5	25
6	6	33.86	6	36
7	4	33.90	7	49
8	5	34.15	8	64
9	3	34.90	9	81

- Calculate the rank correlation coefficient between years of competitive running and a runner's finish time in the 10-kilometer race.
- Do the data provide evidence to indicate a significant rank correlation between the two variables? Test using $\alpha = .05$.

$$(a) \sum d_i^2 = 208.5, n=9$$

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2-1)} = 1 - \frac{6(208.5)}{9(9^2-1)} = -0.7375$$

負向等級相關

經驗 ↑ : time ↓

$$(b) \begin{array}{l} H_0: \rho_s = 0 \\ H_a: \rho_s \neq 0 \end{array} \quad \text{雙尾} \quad \alpha = \frac{0.05}{2} = 0.025$$

$$n=9, \alpha=0.05$$

$$\text{reject } \Rightarrow r_s < -0.7 \text{ or } r_s > 0.7$$

$r_{9,0.025}$

$$\therefore |r_s| = 0.7375 > 0.7 \quad \therefore \text{reject } H_0$$

sufficient evidence to indicate

競速起步時間，完賽時間增加 rank correlation.